

Aaron Leevord Fernandes

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EDUCATION

Indiana University Bloomington

Master of Science in Intelligent Systems Engineering - Computer Engineering

Aug 2023 - May 2025

GPA: 3.9/4.0

Goa University

Bachelor of Engineering - Electronics and Telecommunication

Aug 2017 - Jul 2021

GPA: 3.7/4.0

EXPERIENCE

Machine Learning Engineer, Image Processing, Computer Vision and Optics (Leung Research Group) Apr 2024 – Present

Biomedical Video Segmentation Project

- Designed deep learning **CNN-GRU and Transformers** based architectures in Pytorch for Optical Coherence Tomography (OCT) **unsupervised video segmentation** and **spatio-temporal analysis** to enable virtual histology (tissue analysis)
- Combined **Fourier analysis**, and deep learning, providing 50% better details than previous works.
- Solved tissue drift and warping during imaging by classical **affine registration** methods (Dipy) and voxel morph inspired U-Net based architectures for regularized **diffeomorphic registration** decreasing MSE by 10%.

Metasurfaces: High-Q Biomedical sensing Project

- Built **High Performance Computing** (Slurm, MPI) accelerated Lumerical FDTD simulation pipelines for Meta-Surface design and **synthetic data generation** improving generation speeds by 80%.
- Engineered **U-Net** based FDTD solver **surrogate models** for forward and inverse metasurface design in **tensorflow**, reducing simulation run-time by 70% while maintaining 95% accuracy, achieving up to 15% gains over prior benchmarks.

Software Engineer, Machine Learning (Indiana University)

Jan 2025 - May 2025

Multi Target Path Planning and Routing for Robotic Environments Project

- Identified limitations of value-based methods (e.g., DQN: 20% success) in stochastic, multi-target routing environments.
- Implemented and benchmarked **policy-gradient** algorithms (**REINFORCE, PPO, TRPO, SAC, A2C**) in PyTorch across custom tabular path-planning tasks with moving targets and obstacles; **PPO** achieved up to 90% success.
- Addressed **sparse reward problem** by integrating **Goal-Conditioned** Actor-Critic with **Stein Variational Goal Generation** and **Hindsight Experience Replay (HER)**, enabling successful completion of complex multi-goal tasks **previously unsolvable** under standard RL setups.

Software Engineer (Tata Consultancy Services - HDFC Bank)

Jan 2022 - Jan 2023

- Developed robust financial transaction systems using **Java, Spring and Hibernate**, improving processing efficiency
- Engineered SOAP/HTTP APIs and **Microservices**, enhancing system interoperability and reducing latency by 10-15%
- Streamlined transactional workflows, ensuring data consistency and seamless integration, cutting reconciliation errors 7-15%

TECHNICAL SKILLS

Machine Learning / AI: PyTorch, TensorFlow, Keras, JAX, Hugging Face Transformers, LangChain, LlamaIndex, LangGraph, spaCy, NLTK, LLMs (BERT, GPT, RAG), Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, FAISS, Pinecone, Weaviate, Chroma

Systems / Cloud / DevOps: AWS (SageMaker, EC2, Lambda, S3, RDS, CloudFormation), GCP (Functions, BigQuery, Pub/Sub, Memorystore), Docker, Kubernetes, Spark, Hadoop, CI/CD, Microservices, Serverless

Programming Languages: Python, Java, C, C++, JavaScript (Node.js/TypeScript), SQL

Full-Stack Development: React.js, Next.js, Node.js, Spring Boot, Hibernate, PostgreSQL, MongoDB, Redis, REST, GraphQL, Neo4j

Certifications: AWS Developer Associate, AWS Cloud Practitioner

PROJECTS

Booking Management System | Next.js, Python, FastAPI, MySQL

June 24 - Aug 2024

- Built **10+ key features** (secure login, email, chat, Stripe) with unit tests, reducing production bugs by 40%.
- Implemented **RESTful APIs** for bookings and payments, integrating Stripe and optimizing frontend-backend communication, improving UX and transaction success rates.

Grounded Theory Analysis Automation App | Electron.js, React.js, LangChain — NICC, Brussels

Oct 2024 - Dec 2024

- Developed a **cross-platform** desktop application for interactive relationship visualization and theory extraction using **Knowledge Graphs (Neo4j)**, LLM-powered NLP pipelines (**LangChain and LlamaIndex**) to process qualitative data.
- Targeted the 'coding' phase of GT framework, generating 90% of human codes and cutting analysis time from months to days.

Serverless Matrix Multiplication Using Map-Reduce | Google Cloud (GCP)

Aug 2024 - Sept 2024

- Formulated a serverless **Map-Reduce** matrix multiplication framework on **Google Cloud (Cloud Functions, Pub/Sub, Memorystore)** enabling scalable processing of matrices up to 10 GB, achieving a 25% efficiency gain by addressing key bottlenecks.

Efficient Model-Based Deep Reinforcement Learning With Predictive Control

Oct 2024 - Dec 2024

- For better sample efficiency in RL methods, proposed an MPC based RL framework with **system Identification (World Model)** and **Value Iteration**, converging in 200 episodes (best case) versus 1000+ for model-free methods.
- Learned dynamics model allowed the use of **Cross Entropy Method** for trajectory optimization and planning using the learning Model Network (Delta prediction network), increasing sample efficiency. (80% faster convergence)

Vision Transformer-Based Unsupervised Anomaly Detection on Industrial Dataset

Dec 2024 - Jan 2025

- Constructed a Vision Transformer-based autoencoder using **ViT** and **capsule networks** with **Gaussian Density Mixture Model**, attaining **20% higher AUROC** than CNN-based methods on the BeanTech AD dataset